The following text is adapted from Virginia Woolf's 1919 novel *Night and Day*. Katharine is the granddaughter of a celebrated poet.

> [Katharine's] descent from [a celebrated poet] was no surprise to her, but matter for satisfaction, until, as the years wore on, certain drawbacks made themselves very manifest. Perhaps it is a little depressing to inherit not lands but an example of intellectual virtue; perhaps the conclusiveness of a great ancestor is a little discouraging to those who run the risk of comparison with him.

3 Mark for Review

As used in the text, what does the word "manifest" most nearly mean?

(A) Possible

(B) Evident

(C) Complex

(D) Particular

@DSATuz

The discoverers of the minor planet 2731 Cucula named it after the animal genus that includes cuckoos. Most of the recently discovered minor planets, however, are given only an identification number, a trend that can be _____ to there being over 500,000 such bodies known at present, making it nearly impossible to find a unique name for each.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) attributed

(B) referred

(C) alluded

(D) entrusted

@DSAT.uz

Many visitors to Yellowstone National Park need airlines, car rentals, hotels, and restaurants during their visits. These services generate carbon emissions. Emily Wilkins and Jordan Smith found that most of the environmentally harmful emissions come from travel services to and from the park. Wilkins and Smith found that policymakers could help reduce these emissions by encouraging people to visit state or national parks closer to their homes to cut down on their travel.

5 Mark for Review

Which choice best states the function of the underlined sentence in the text as a whole?

(A) It presents a harm that is unique to the location described earlier in the text.

(B) It indicates that policymakers have been working on a certain problem for a long time.

(C) It suggests a course of action to address the problem described earlier in the text.

(D) It rephrases the question raised in the first sentence of the text.

@DSATuz

Though John Crowley, author of *Flint and Mirror*, is perhaps not as well known as the most commercially successful American writers of the past fifty years, his work has had several influential champions, including the poet James Merrill and the literary critic Harold Bloom. In his afterword to Crowley's book *Little, Big*, Bloom praises how the novel adroitly blends what playwright Friedrich Schiller termed the naive and sentimental modes—while Schiller thought works could be classified as either naive (seeking to describe reality) or sentimental (seeking to develop ideas), *Little, Big* fuses both modes of writing.

6 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

(A) It gives a reason why a critic appreciates a certain book.

(B) It helps clarify the distinction between two categories.

(C) It compares the work of a writer with that of a poet who admired him.

(D) It contrasts the opinions of two critics.

@DSAT.uz

The following text is adapted from Anthony Trollope's 1855 novel *The Warden*. The narrator is describing a rectory, the residence of a clergyperson.

> Let us observe the well-furnished breakfast-parlour at Plumstead Episcopi, and the comfortable air of all the belongings of the rectory. Comfortable they certainly were, but neither gorgeous nor even grand; indeed, considering the money that had been spent there, the eye and taste might have been better served; there was an air of heaviness about the rooms which might have been avoided without any sacrifice of propriety; colours might have been better chosen and lights more perfectly diffused; but perhaps in doing so the thorough clerical aspect of the whole might have been somewhat marred.

7 Mark for Review

Which choice best states the function of the underlined portion in the text as a whole?

(A) It offers a potential explanation for the furnishing choices the narrator critiques.

(B) It emphasizes that the narrator finds it inappropriate for a clergyperson to have such grand furnishings.

(C) It signals how the inexpensive-looking ornaments in the rectory could be enhanced.

(D) It concedes that there is a lack of attention to the impression the rectory makes on people.

@DSATuz

Text 1

Ollantay is a play in Quechua, an Indigenous language in South America. The play portrays life in the Inca Empire before Spain invaded in the 1500s. Yet the oldest known text of the play is from 1770, and in many places its writing style resembles the writing style of Spanish plays from the 1700s. Thus, *Ollantay* was probably created in the late 1700s by someone who knew much about the Inca past.

Text 2

Ollantay includes details of Inca society that a writer in the 1700s wouldn't have known about. Also, the play's structure matches descriptions of the structure of Inca drama from the 1500s. The obvious explanation is that *Ollantay* itself is an Inca work from the 1500s. It could have been performed continuously from the 1500s to the 1700s. By the time it was written down in 1770, it could have easily been influenced by later Spanish plays.

8 Mark for Review

Based on the texts, what do the author of Text 1 and the author of Text 2 mainly disagree about?

(A) What inspired a Spanish writer in the 1700s to write a play about the Inca Empire

(B) Whether *Ollantay* is an Inca play from the 1500s or was instead created much later, in the 1700s

(C) How the Inca author of *Ollantay* knew so much about plays from Spain

(D) Why Inca plays became more popular in the 1700s than they had been when they were first performed in the 1500s

@DSATuz

In a period of great prosperity during his reign, Egyptian pharaoh Amenemhat III (circa 1800 BCE) had a pyramid with a massive substructure built at Hawara. The underground complex was largely dismantled in antiquity, but accounts of its splendor remain, including one from Greek historian Herodotus, who claimed to have seen it. In his *Histories* (circa 425 BCE), Herodotus writes of 3,000 chambers adorned with carved figures, a dozen courts flanked by white stone pillars, and many winding passages in the complex, characterizing it as a great labyrinth and "an unending marvel."

9 Mark for Review

According to the text, what information did Herodotus include in his *Histories*?

(A) An account of the magnificence of the complex that once stood beneath the pyramid at Hawara

(B) A record of how long it took for the pyramid and complex to be constructed at Hawara

(C) A discussion of how building the pyramid and complex at Hawara led to a period of prosperity

(D) An explanation of how Herodotus was able to arrange to see the pyramid and complex at Hawara

@DSAT.uz

Dubautia carrii is among the twenty-eight species of silversword plants found only on the Hawaiian archipelago that collectively illustrate the process of adaptive radiation, or the rapid diversification of an ancestral species into different, related species. Each silversword species is physically distinct, with mature plant forms ranging from trees and shrubs to vines. However, they all descended from a common tarweed plant species, with their unique physical characteristics emerging as they adapted to the archipelago's many specific habitats over time.

10 Mark for Review

Which choice best states the main idea of the text?

(A) All silverswords that grow on the Hawaiian archipelago have similar physical characteristics.

(B) The Hawaiian archipelago exhibits many distinct habitats and species.

(C) Most plants that grow on the Hawaiian archipelago are descendants of a single founder species.

(D) Silverswords are good examples of adaptive radiation on the Hawaiian archipelago.

@DSATuz

Total Areas of Five Tribal Nations around the United States

Tribal nation	Location	Area (square miles)
White Mountain Apache Tribe	Arizona	2,631
Crow Tribe	Montana	3,606
Leech Lake Band of Ojibwe	Minnesota	1,311
Chickasaw Nation	Oklahoma	7,648
Cheyenne River Sioux Tribe	South Dakota	4,419

A citizen of the Leech Lake Band of Ojibwe is a student in a geography class. The student wants to compare the total area covered by her tribal nation in Minnesota with the total areas covered by certain tribal nations in other states. Looking at the table, she finds that the area of the Leech Lake Band of Ojibwe is 1,311 square miles, while the area of the White Mountain Apache Tribe in Arizona is _____

11 Mark for Review

Which choice most effectively uses data from the table to complete the comparison?

(A) 7,648 square miles.

(B) 2,631 square miles.

(C) 4,419 square miles.

(D) 3,606 square miles.

@DSATuz

As of 2023, Turkey has 1,015 kilometers of high-speed rail lines in operation and 614 kilometers under construction; in Germany, 1,630 kilometers are in operation with 3,262 kilometers under construction. In contrast, the United States has a relatively low amount of high-speed rail for its size: only 735 kilometers of high-speed rail in operation and 1,789 kilometers under construction. Many US transportation experts argue that this should change, claiming that high-speed rail is more cost-effective than highway systems are.

12 Mark for Review

Which finding, if true, would most directly support the US transportation experts' claim?

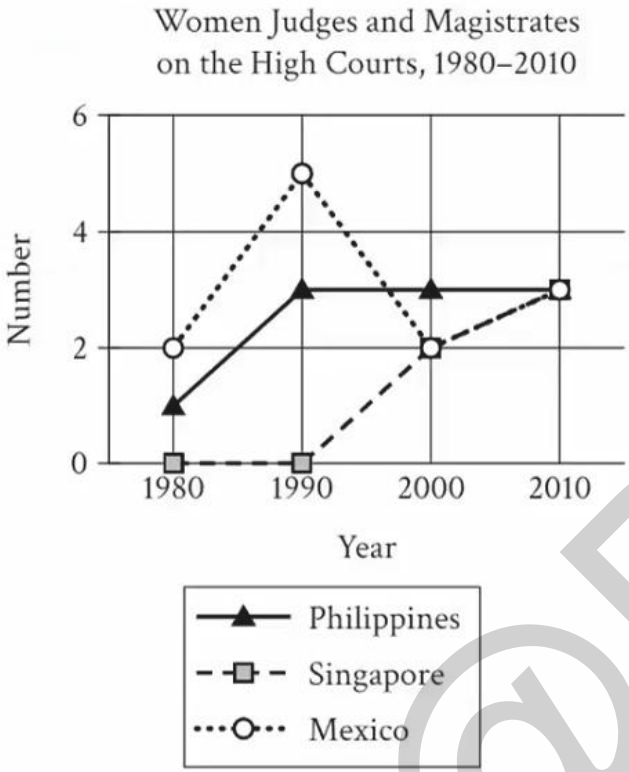
(A) In most locations in the US, increasing highway capacity is more expensive relative to the volume of traffic that can be accommodated than increasing high-speed rail capacity is.

(B) In much of the US, the existing rail infrastructure is so old that it would have to be replaced entirely in order to accommodate high-speed trains.

(C) Building successful high-speed rail infrastructure would require that the US government establish a fund to maintain the rail lines long-term.

(D) Unlike people in the United States, people in Turkey and Germany are less interested in traveling by highway than they are in traveling by high-speed rail.

@DSATuz



A report from an international organization that monitors the numbers of women serving as judges or magistrates on various nations' highest courts, such as the Supreme Court of Justice in Mexico and the Supreme

13 Mark for Review

Which choice most effectively uses data from the graph to complete the example?

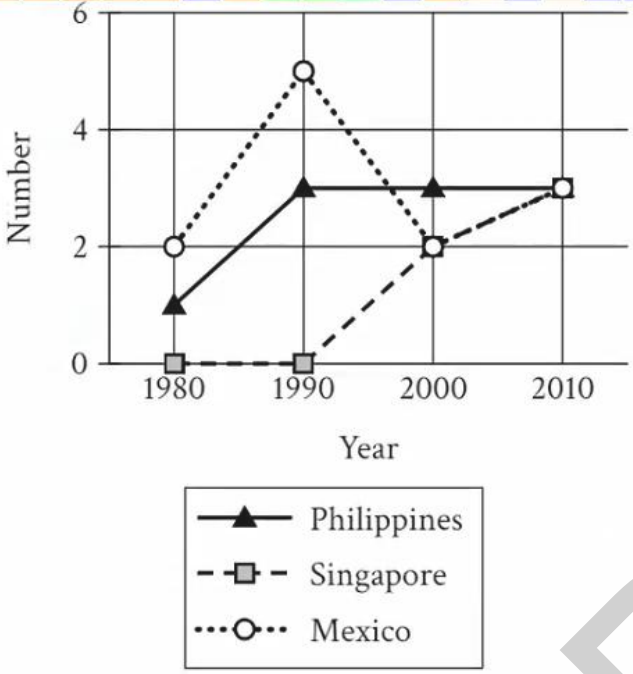
(A) in 2010, Singapore had 3 women on its high courts, the Philippines had 3, and Mexico had 3.

(B) neither Singapore nor the Philippines saw a reduction in the women on their high courts in any of the years shown on the graph, but Mexico did after 1990.

(C) the increase in the number of women on the high courts from 1980 to 2010 in Singapore was greater than that in either the Philippines or Mexico.

(D) Mexico had 2 women on its high courts in 1980 and 3 in 2010.

@DSAT.uz



A report from an international organization that monitors the numbers of women serving as judges or magistrates on various nations' highest courts, such as the Supreme Court of Justice in Mexico and the Supreme Court in the Philippines, found that the overall trend is toward more women serving on the high courts in 2010 than in 1980. For example, none of the countries in the graph had more than 2 women in these positions in 1980, but _____

13 Mark for Review

Which choice most effectively uses data from the graph to complete the example?

(A) in 2010, Singapore had 3 women on its high courts, the Philippines had 3, and Mexico had 3.

(B) neither Singapore nor the Philippines saw a reduction in the women on their high courts in any of the years shown on the graph, but Mexico did after 1990.

(C) the increase in the number of women on the high courts from 1980 to 2010 in Singapore was greater than that in either the Philippines or Mexico.

(D) Mexico had 2 women on its high courts in 1980 and 3 in 2010.

@DSAT.uz

Careful planning was essential for the success of the Deep Impact probe, which conducted a flyby of comet 9P/Tempel and then traveled to comet 103P/Hartley. The CONTOUR probe mission seemed similarly well planned, but engineers lost communications with the probe before it could achieve the mission goal of visiting comet 73P/Schwassmann-Wachmann. This shows that _____

14 Mark for Review

Which choice most logically completes the text?

(A) the Deep Impact was more successful than the CONTOUR because it had fewer tasks to accomplish.

(B) having multiple mission goals ensures that a probe will accomplish at least one of its tasks.

(C) the CONTOUR would likely have accomplished its tasks if it had been sent somewhere other than comet 73P/Schwassmann-Wachmann.

(D) careful planning alone does not guarantee that a probe will accomplish its mission.

@DSAT.uz

Dermatological researchers and geneticists make regular use of representative cross-sections of hair fibers, but such a cross-section can be obtained only if the strand can be cut perpendicular to its length without distorting the cross-sectional shape, which necessitates embedding the hair in a stabilizing medium, often resin, before sectioning it. Resin is effective but slow due to its long curing time, which led Tina Lasisi and colleagues to try sheets of low-melt-point plastic as a stabilizer. They found that these sheets match resin's utility as a stabilizer while allowing for near-instantaneous embedding of hair, thereby demonstrating that using low-melt-point plastic sheeting rather than resin can enhance _____

15 Mark for Review

Which choice most logically completes the text?

(A) an embedded hair's cross-sectional shape without reducing the researchers' likelihood of obtaining perpendicular cuts.

(B) the efficiency of preparatory steps before sectioning a hair without compromising the hair's cross-sectional shape.

(C) the rate at which hairs can be embedded before sectioning because no stabilizing agent is used.

(D) the researchers' likelihood of obtaining perpendicular cuts while simplifying the process of preparing a hair for sectioning.

@DSATuz

In the 1920s and 1930s, the singer Frank "Machito" Grillo performed with Cuba's most popular dance orchestras. However, the fusion of Latin music and American jazz that Machito innovated while performing in New York City is what has _____

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) earned him his enduring fame?

(B) his enduring fame earned him?

(C) his enduring fame earned him.

(D) earned him his enduring fame.

@DSA_Tuz

Before the Spanish arrived in the 1400s, the people of La Gomera, one of the Canary Islands, would use a whistled version of their native language to "speak" to each other from far away. Amazingly, under Spanish rule, the _____ adapted their whistling skills to whistle to each other in Spanish.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) island's residents

(B) islands resident's

(C) island's resident's

(D) islands residents'

@DSATuz

Assembling _____ by hand and in painstaking detail, Nigerian American artist Njideka Akunyili Crosby works at a methodical pace when completing her massive collage-based paintings.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) it

(B) that

(C) her

(D) them

@DSATuz

Like almost every country in the world, Britain has been allocated an international dialing prefix to help route incoming international calls. First assigned to Britain in the 1960s Red Book, a directory published by a United Nations telecommunications agency, _____

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) international direct dialing was not widely available until the 1970s, so initially, only switchboard operators used the prefix.

(B) not until the 1970s did international direct dialing become widely available, meaning that, initially, only switchboard operators used the prefix.

(C) initially, only switchboard operators used the prefix, as international direct dialing was not widely available until the 1970s.

(D) the prefix was used by only switchboard operators initially, as international direct dialing was not widely available until the 1970s.

@DSATuz

Artist Yto Barrada's exhibition *Ways to Baffle the Wind* incorporates sculptures, textiles, and films. Barrada's pieces, utilizing elements as disparate as plant-dyed fabrics, wire crab traps filled with stones, and cotton balls dangling above a fan, _____ the ways humans attempt to organize and regulate nature.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) has explored

(B) explore

(C) explores

(D) exploring

@DSATuz

A team led by Portuguese researcher Isabel C.F.R. Ferreira found that many species of mushrooms contain chemicals called phenolic compounds, such as cinnamic acid and catechin. _____ Ferreira detected cinnamic acid in *Agaricus blazei* mushrooms and catechin in *Lentinus edodes* mushrooms.

21 Mark for Review

Which choice completes the text with the most logical transition?

(A) For this reason,

(B) However,

(C) Nevertheless,

(D) For example,

@DSATuz

Artist and writer Kameelah Janan Rasheed wanted to demonstrate the value of textual revision and to encourage audiences to embrace it as a sign of learning. _____ she was transparent about the revision process for her 2019 book *No New Theories*, adding annotations to the original text to indicate where her ideas had shifted or deepened.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) Thus,

(B) Nevertheless,

(C) Instead,

(D) Later,

@DSATuz

In a hypothetical family tree, it is common for a parent in the first generation to exhibit a given trait, such as dimples, and for none of that parent's children (second generation) to inherit it. _____ the grandchildren (third generation) could inherit dimples or another such trait from the grandparent.

23 Mark for Review

Which choice completes the text with the most logical transition?

(A) Nonetheless,

(B) Specifically,

(C) Besides,

(D) In addition,

@DSATuz

Teams from 32 countries gathered in Brazil in 2014 for the World Cup of men's soccer. The event, which is held every four years, now features teams from all over the world. The World Cup used to be much smaller and more limited geographically, _____ The 1934 event in Italy included only 16 teams, most from Europe and the Americas.

24 Mark for Review

Which choice completes the text with the most logical transition?

(A) indeed.

(B) though.

(C) for example.

(D) moreover.

@DSATuz

While researching a topic, a student has taken the following notes:

- Most of the plant and bird species in Oahu, Hawaii, are non-native.
- In a 2019 study, researchers wanted to know what role non-native birds play in dispersing plant seeds in Oahu.
- Researchers catalogued plant seeds found in fecal samples from non-native birds.
- *Psydrax odorata*, a flowering shrub, was one of fifteen native species catalogued.
- *Hedychium gardnerianum*, an herbaceous vine, was one of twenty-nine non-native species catalogued.
- Researchers concluded that non-native birds play a vital role in dispersing the seeds of native and non-native plants.

25 Mark for Review

The student wants to explain what *Psydrax odorata* is. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) In a 2019 study, researchers catalogued seeds from *Psydrax odorata*.

(B) *Psydrax odorata* is a flowering shrub species that is native to Oahu, Hawaii.

(C) Most plant species found in Oahu, Hawaii, like *Hedychium gardnerianum*, are non-native.

(D) Seeds from *Psydrax odorata* were found by researchers in the feces of non-native birds.

@DSATuz

While researching a topic, a student has taken the following notes:

- *The Future of Nostalgia* is a scholarly book by literary theorist Svetlana Boym.
- The book provides a multifaceted exploration of the concept of nostalgia.
- Chapter 4 defines a form of nostalgia interested in re-creating the past.
- Chapter 5 explores the Soviet-style restaurant the Nostalgija Snack Bar in Ljubljana.
- In chapter 5, Boym writes, "[The Nostalgija Snack Bar] is a new kind space that plays with the past and the present."

26 Mark for Review

The student wants to provide a quotation from chapter 5. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Svetlana Boym defines a form of nostalgia interested in re-creating the past in chapter 4 of her book.

(B) In a discussion of the Soviet-style restaurant the Nostalgija Snack Bar in Ljubljana, Svetlana Boym writes, "[The Nostalgija Snack Bar] is a new kind space that plays with the past and the present."

(C) Svetlana Boym's *The Future of Nostalgia* provides a multifaceted exploration of nostalgia.

(D) After defining a form of nostalgia interested in re-creating the past, Svetlana Boym goes on to explore the Soviet-style restaurant the Nostalgija Snack Bar in Ljubljana.

@DSATuz

While researching a topic, a student has taken the following notes:

- 1905: US astronomer Percival Lowell begins searching beyond Neptune for a mysterious planet he calls "Planet X."
- 1916: Lowell dies but leaves money to upgrade his observatory.
- 1929: A new telescope is built at the observatory.
- February 1930: Observatory assistant Clyde Tombaugh records the first photographic evidence of Planet X.
- March 1930: Astronomers rename Planet X, calling it Pluto to honor Lowell (the first two letters are Lowell's initials, PL).

27 Mark for Review

The student wants to specify Percival Lowell's contributions to the discovery of Pluto. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Astronomers renamed Planet X in 1930 to honor Lowell for his contributions to the discovery of Pluto.

(B) After a new telescope was built at Lowell's observatory in 1929, Clyde Tombaugh recorded the first photographic evidence of Planet X, later renamed Pluto.

(C) In 1930, Lowell became the first person to record photographic evidence of Pluto, known at the time as Planet X.

(D) Lowell began the search for Pluto in 1905 and later left money to upgrade the observatory where it was discovered in 1930.

@DSATuz

Scholars long thought that the initial spread of silk beyond China occurred in the second century CE, but this view has been _____ by new archaeological evidence from South Asia that reveals that the people of the Indus Civilization made use of silk at least 1,000 years earlier.

1 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) investigated

(B) contradicted

(C) misinterpreted

(D) anticipated

@DSAT.uz

In a 2022 study, Nicolás M. Morato and colleagues attempted to resolve a long-standing paradox: although water is critical to most reactions that enable life, during the earliest stages of Earth's history—when most of the planet's surface was covered by oceans—the presence of water likely _____ the formation of proteins and other biomolecules that are found in all living organisms.

2 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) exploited

(B) imitated

(C) hampered

(D) reflected

@DSATuz

A team of microbiologists measured surface temperatures of various wild mushrooms, including *Amanita muscaria* and a species from the genus *Thelephora*. The temperatures ranged from 1.4 to 5.9°C lower than the temperature of the surrounding air, a difference the team _____ to water evaporating from the mushrooms and thus cooling them.

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) conceded

(B) prescribed

(C) attributed

(D) amended

@DSATuz

Despite stated claims of the _____ of the research, much analysis of income inequality performed in the 2010s suffered from a myopic focus on a few countries in North America and Western Europe, partly due to limited data availability. Researchers would later remedy this shortcoming after gaining new access to administrative records located in nations in Africa, such as Ivory Coast, and Eastern Europe, such as Poland.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) novelty

(B) immutability

(C) exhaustiveness

(D) pervasiveness

@DSAT.uz

The following text is from Yung Wing's 1909 memoir *My Life in China and America*. Yung Wing was the first person from China to graduate from a US university.

> Little did I realize when in 1845 I wrote, while in the Morrison School, a composition on "An Imaginary Voyage to New York and up the Hudson," that I was to see New York in reality. This incident leads me to the reflection that sometimes our imagination foreshadows what lies uppermost in our minds and brings possibilities within the sphere of realities.

5 Mark for Review

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It foreshadows Yung's eventual success as a writer.

(B) It indicates Yung's unwillingness to distinguish between reality and fantasy as a child.

(C) It illustrates the sense of adventure that Yung developed as a child.

(D) It describes an event in Yung's life that exemplifies a phenomenon.

@DSATUZ

The following text is from Jerome K. Jerome's 1889 novel *Three Men in a Boat (To Say Nothing of the Dog)*. The narrator and two friends are taking a boat down the River Thames in England.

> In a boat, I have always noticed that it is the fixed idea of each member of the crew that he is doing everything. Harris's notion was, that it was he alone who had been working, and that both George and I had been imposing upon him. George, on the other hand, ridiculed the idea of Harris's having done anything more than eat and sleep, and had a cast-iron opinion that it was he—George himself—who had done all the labour worth speaking of.

6 Mark for Review

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It establishes Harris's feelings about the narrator's idea of how boat trips usually proceed.

(B) It offers Harris's belief as a specific example of a trend the narrator has observed while boating.

(C) It suggests that Harris is focused on helping the group navigate a challenge.

(D) It demonstrates that Harris finds his experiences on the boat to be confusing.

@DSATuz

Driven to sell as many paintings as possible, Alfred Hair, an influential figure among the landscape artists known as the Florida Highwaymen, pioneered "fast painting," a technique (which in part involved swift applications of paint) that many Highwaymen, including Livingston Roberts, adopted. To conclude that this approach accounts for the ethereal qualities now synonymous with the Highwaymen aesthetic is tempting but inaccurate, as Hair's methods weren't universally practiced by his affiliates: George Buckner, for example, painted with greater deliberateness but achieved the same effects.

7 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

(A) It details evidence that contradicts a claim mentioned earlier in the text about a long-standing disagreement within a group of artists.

(B) It considers and rebuts an interpretation of the effect of a painting technique mentioned earlier in the text on the perception of work by a group of artists.

(C) It establishes a contrast between the aesthetic qualities of works by artists who were central to a movement introduced earlier in the text and those of an artist who was more peripheral to that movement.

(D) It explains how an artist mentioned earlier in the text developed a distinctive style by adapting a particular approach to painting originated by his colleagues.

@DSATuz

Studies of ocean wave breaking have predominantly focused on traveling waves (those propagating along the horizontal plane), so Mark McAllister et al. utilized a circular wave tank to produce and study spike waves, axisymmetric standing waves that can erupt vertically when traveling waves propagating in opposing directions intersect. Traveling waves break when wave steepness (height-to-length ratio) passes a critical threshold; breaking thus constrains wave height. McAllister et al. found that spike waves can exceed that constraint, as other factors than just steepness (e.g., jet stability and cavity shape) mediate spike-wave breaking.

8 Mark for Review

Which choice best states the main idea of the text?

(A) McAllister et al. suggest that spike waves can form when traveling waves propagating in opposing directions intersect and that spike waves tend to be higher than traveling waves.

(B) The process of breaking limits the height of traveling waves, but the study by McAllister et al. suggests that spike waves can exceed those limits if their height-to-length ratio reaches a critical threshold.

(C) The study by McAllister et al. suggests that when traveling waves intersect in specific ways, the resulting wave may be higher than would be expected based on the properties of traveling waves.

(D) Previous studies have suggested that steepness mediates breaking in traveling waves, but the study by McAllister et al. shows that jet stability and cavity shape may also influence breaking in such waves.

@DSATuz

Optimal foraging theory (OFT) holds that animals' foraging behaviors reflect cost-benefit trade-offs that vary by species and with dynamic ecological circumstances. One such circumstance is lunar intensity, which Joel Brown and colleagues found to be negatively associated with foraging by Arizona pocket mice but Patricia C. Wright found to be positively associated with foraging by three-striped night monkeys. This discrepancy is explicable in terms of OFT: the monkeys' greater reliance on vision means that higher lunar intensity benefits them more than it benefits the mice.

9 Mark for Review

Which choice best describes the finding made by Wright, as presented in the text?

(A) During periods of high lunar intensity, three-striped night monkeys increase their reliance on vision.

(B) As lunar intensity increases, Arizona pocket mice reduce their foraging activity.

(C) As lunar intensity increases, three-striped night monkeys increase their foraging activity.

(D) Increased lunar intensity is more beneficial for three-striped night monkeys than it is for Arizona pocket mice.

@DSATuz

Nautilids are marine mollusks that begin growing their shells before emerging from their eggs and continue to add shell segments throughout their lifetimes. The walls between their shells' chambers are called septa, and the deeper the water in which a septum forms, the greater the concentration of the isotope oxygen-18 the septum will contain. Since temperature falls as depth increases, if the area of ocean a nautilid inhabits is known, this isotopic signature can reveal the temperature at which its septa formed. Paleontologist Amane Tajika and colleagues examined each of the septa in two nautilid shells and concluded that septum sample M23 formed at a temperature of 14.0°C whereas sample F02 formed at 20.8°C.

10 Mark for Review

Which finding, if true, would most directly weaken the researchers' conclusion?

(A) At a depth of 140 meters in the nautilids' habitat, the water is 20.8°C and the concentration of oxygen-18 is equal to that in sample F02.

(B) The concentration of oxygen-18 in F02 is higher than that in M23.

(C) The concentration of oxygen-18 in F02 is lower than that in M23.

(D) At a depth of 385 meters in the nautilids' habitat, the water is 14.0°C and the concentration of oxygen-18 is equal to that in sample M23.

@DSAT.uz

In Diné (Navajo) culture, *ikaah* (sandpaintings) are created and then erased as part of sacred healing ceremonies lasting no more than a few days, but Diné *hataałii* (chanter, healer, cultural guide) Fred Stevens developed fixatives to preserve desacralized sandpaintings. While on a US-sponsored cultural ambassadorial trip in Europe and the Americas in the 1960s, Stevens produced several such sandpaintings and gave them to cultural institutions. This may seem in tension with the role of cultural ambassador—how could static objects authentically represent an inherently ephemeral and dynamic practice?—but such a view is itself overly object-focused and neglects how Stevens strove to convey exactly those characteristics of *ikaah* as a cultural practice.

11 Mark for Review

Which quotation from an art historian would most directly support the claim made in the text?

(A) "Stevens's ambassadorial sandpaintings are best understood not as self-contained objects but as reminders of the public creations of the sandpaintings, during which Stevens conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process."

(B) "While Stevens's ambassadorial sandpaintings are undoubtedly educative for audiences that have little exposure to Diné culture, they should not be confused with authentic *ikaah*, which cannot be extricated from a practice that is intentionally and necessarily transitory nor condensed into a single persistent object."

(C) "The most compelling way to reconcile the apparent tension between the temporary and performative nature of *ikaah* and the persistent and static nature of Stevens's ambassadorial sandpaintings is to recognize that Stevens was an ambassador

@DSATuz

In Diné (Navajo) culture, *ikaah* (sandpaintings) are created and then erased as part of sacred healing ceremonies lasting no more than a few days, but Diné *hataałii* (chanter, healer, cultural guide) Fred Stevens developed fixatives to preserve desacralized sandpaintings. While on a US-sponsored cultural ambassadorial trip in Europe and the Americas in the 1960s, Stevens produced several such sandpaintings and gave them to cultural institutions. This may seem in tension with the role of cultural ambassador—how could static objects authentically represent an inherently ephemeral and dynamic practice?—but such a view is itself overly object-focused and neglects how Stevens strove to convey exactly those characteristics of *ikaah* as a cultural practice.

track his movements and subtle shifts in the sand throughout the process."

(B) "While Stevens's ambassadorial sandpaintings are undoubtedly educative for audiences that have little exposure to Diné culture, they should not be confused with authentic *ikaah*, which cannot be extricated from a practice that is intentionally and necessarily transitory nor condensed into a single persistent object."

(C) "The most compelling way to reconcile the apparent tension between the temporary and performative nature of *ikaah* and the persistent and static nature of Stevens's ambassadorial sandpaintings is to recognize that Stevens was an ambassador not only of Diné culture but of a US art culture that tended to value permanent works over ephemeral ones."

(D) "It is important to recall Stevens's own actions as a gift-giver—a temporary and performative role—of the sandpaintings, for by taking those actions, Stevens publicly enacted the transmission of knowledge of both traditional *ikaah* practices and cutting-edge chemistry (in the form of the preservative additives), a hybrid that reflects the dynamism of Diné culture."

@DSATUZ

Percentage of Jumping Spiders That Attacked Termites Wearing a Certain Cape Type, by Cape Type First Looked At

Termite cape type the spiders looked at first	Percent of spider attacks on termites with solid black capes	Percent of spider attacks on termites with solid white capes	Percent of spider attacks on termites with black-and-white striped capes
solid black cape	60%	26%	13%
solid white cape	14%	86%	0%
black-and-white striped cape	25%	50%	25%

Some animals evade predation with the help of contrasting markings that signal toxicity. Lisa Taylor and team studied how jumping spiders behave toward termites (their regular prey) to which solid black, solid white, or black-and-white striped (i.e., contrasting) capes were affixed. Each spider

12 Mark for Review

Which choice best describes data from the table that support Taylor and team's conclusion?

(A) Half of the spiders that first looked at a termite wearing a black-and-white striped cape attacked a termite wearing a solid white cape, and the other half of those spiders were just as likely to attack a termite wearing a black-and-white striped cape as they were to attack a termite wearing a solid black cape.

(B) For each type of cape, a majority of the spiders that first looked at a termite wearing that type of cape attacked a termite wearing that same type of cape.

(C) Regardless of the type of cape worn by termites the spiders first looked at, the large majority of spider attacks were on termites wearing solid-color capes.

(D) None of the spiders that first looked at a termite wearing a solid white cape attacked a termite wearing a black-and-white striped

@DSATuz

type the spiders looked at first	spider attacks on termites with solid black capes	spider attacks on termites with solid white capes	attacks on termites with black-and-white striped capes
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Some animals evade predation with the help of contrasting markings that signal toxicity. Lisa Taylor and team studied how jumping spiders behave toward termites (their regular prey) to which solid black, solid white, or black-and-white striped (i.e., contrasting) capes were affixed. Each spider was placed in a clear container in the center of a circular dish in which six termites, two with each of the three cape types, could be seen. Noting which termites the spiders first looked at and which termites they ultimately targeted, the researchers concluded that contrasting markings were a deterrent.

12 Mark for Review

Which choice best describes data from the table that support Taylor and team's conclusion?

(A) Half of the spiders that first looked at a termite wearing a black-and-white striped cape attacked a termite wearing a solid white cape, and the other half of those spiders were just as likely to attack a termite wearing a black-and-white striped cape as they were to attack a termite wearing a solid black cape.

(B) For each type of cape, a majority of the spiders that first looked at a termite wearing that type of cape attacked a termite wearing that same type of cape.

(C) Regardless of the type of cape worn by termites the spiders first looked at, the large majority of spider attacks were on termites wearing solid-color capes.

(D) None of the spiders that first looked at a termite wearing a solid white cape attacked a termite wearing a black-and-white striped cape.

@DSATUZ

Rafael Núñez and colleagues studied how members of the Yupno, an Indigenous group in Papua New Guinea, conceptualize time in both spoken language and gestures. The researchers recorded Yupno speakers explaining certain temporal words and phrases, such as *omo-ropmo bilak*, a past-oriented expression that translates to "a couple of years ago," and coded each speaker's manual gestures. Previous research has found a tendency in many cultures to make temporal distinctions along imagined linear axes: for instance, English speakers often refer to the front/back axis to describe events in time. Some researchers believe this tendency is universal, but Núñez and colleagues claim this is not the case.

13 Mark for Review

Which finding, if true, would most directly support Núñez and colleagues' claim?

(A) Yupno speakers typically use their left hand to make temporal gestures regardless of whether the gestures are past oriented or future oriented.

(B) Some Yupno grammatical structures used when talking about time are also used in English.

(C) Yupno speakers were observed making temporal gestures both indoors and outdoors, though with greater frequency when indoors.

(D) Future-oriented gestures used by Yupno speakers do not, on average, point in the opposite linear direction of past-oriented gestures.

@DSATuz

Whistler waves are low-frequency plasma waves that on Earth are typically generated by lightning. Numerous recordings of whistler waves on Venus have led many scientists to suggest that the planet's atmosphere is host to extensive amounts of lightning, and, in fact, Venusian whistler waves have similar energy signatures to those of whistler waves generated by lightning on Earth. The majority of Venusian whistler wave data come from two spacecraft missions—the Pioneer Venus Orbiter (PVO) and the Venus Express (VEX)—which have included few observations of other phenomena consistent with lightning occurrences (such as flashes of light), leading other scientists to suggest that _____

14 Mark for Review

Which choice most logically completes the text?

(A) there are geophysical characteristics of Venus not shared with Earth that promote the generation of whistler waves.

(B) the purported Venusian whistler waves must actually be some other type of atmospheric activity than whistler waves.

(C) Venusian lightning has properties that make it unlikely to generate whistler waves.

(D) similarities in the energy signatures of Venusian and Earth whistler waves may reflect imprecisions in the PVO and VEX data.

@DSAT.uz

In the world's democracies, proportionally few elected representatives are young adults. While some philosophers argue that certain types of underrepresentation (of women, for example) are unfair in principle and should be remediated via quotas, there is broad agreement that youth underrepresentation is not inherently unfair since it is universal (all people are underrepresented when young), temporary (underrepresented young people become overrepresented older people), and reflective of a legitimate concern (lack of relevant experience). Accordingly, when philosopher Juliana Uhuru Bidadanure advocates for youth quotas for elected representatives, she does so on the basis of their instrumental value for promoting youth interests, arguing that _____

15 Mark for Review

Which choice most logically completes the text?

(A) young adults may lack certain kinds of experience but those experiences are not relevant to the question of whether youth underrepresentation is fair in principle.

(B) age-related quotas would alleviate the inherently unfair overrepresentation of older people that tends to occur in the world's democracies.

(C) youth underrepresentation may be fair in principle but leads to worse outcomes for young adults than could be achieved with more proportionate youth representation.

(D) it does not matter that youth underrepresentation is universal, temporary, and reflective of a legitimate concern since it is inherently unfair.

@DSAT.uz

"Paper Dragons" by James P. Blaylock, *The Ape's Wife and Other Stories* by Caitlin R. Kiernan, and "Confess the Seasons" by Charles L. Grant are all pieces of literature that have won the prestigious World Fantasy _____ the works received the honor in the categories of short story, collection, and novella, respectively.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) Award,

(B) Award;

(C) Award and

(D) Award

@DSATuz

All member nations of the North Atlantic Treaty Organization, or NATO—including Iceland, which joined in 1949; Hungary, which joined in 1999; and Bulgaria, which joined in _____ are committed to NATO's principle of collective defense, each member pledging to defend all others.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) 2004;

(B) 2004,

(C) 2004

(D) 2004—

@DSATuz

Multidisciplinary at its core, _____

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) Diné/Creek artist Anna Tsouhlarakis engages a range of art forms—primarily video, photography, and performance—in her body of work.

(B) video, photography, and performance are among the range of art forms that Diné/Creek artist Anna Tsouhlarakis primarily engages in her body of work.

(C) a range of art forms—primarily video, photography, and performance—are engaged by Diné/Creek artist Anna Tsouhlarakis in her body of work.

(D) Diné/Creek artist Anna Tsouhlarakis's body of work engages a range of art forms—primarily video, photography, and performance.

@DSATuz

At the Chicxulub asteroid impact crater in Mexico, the presence of many fossilized microbacteria, which seem to have thrived there despite the extreme heat that persisted after the Chicxulub impact, _____ claims that bacteria are among the planet's most resilient organisms.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) reinforce

(B) have reinforced

(C) reinforces

(D) are reinforcing

@DSATuz

In Xu Bing's art installation *Book from the Sky*, each one of the thousands of faux Chinese characters—printed in precise columns on the trio of long scrolls that swoop across the ceiling, on the rows of open books that undulate along the floor, and on the paper walls that stand to the sides—_____ no meaning. Invented by Bing, the characters present plausibly and grandiosely, inviting yet punishing those who yearn to decode them.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) hold

(B) holds

(C) have held

(D) were holding

@DSATuz

A loose confederation of cities from eleven modern-day countries, the Hanseatic League was a powerful mercantile alliance that dominated northern European trade between the 13th and 17th centuries. Hanseatic League member _____ was serendipitously located on the Warnow River in Germany, giving Rostock's merchants access to key maritime trade routes.

21 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) city Rostock,

(B) city, Rostock,

(C) city, Rostock

(D) city Rostock

@DSATuz

A staunch supporter of women's voting rights, Wilhelmina Kekelaokalaninui Widemann Dowsett sought to coordinate the efforts of suffragists in her native Hawai'i. _____ in 1912, she founded the National Women's Equal Suffrage Association of Hawai'i, an organization that lobbied for women's voting rights in the US territory.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) Conversely,

(B) To that end,

(C) Alternatively,

(D) In other words,

@DSATuz

In quantum physics, Karen Barad's theory of agential realism—which emphasizes the entanglement, or "intra-action," of human and nonhuman actors—is a foundational rebuke to the dualistic subject-object divide that has undergirded scientific epistemology for centuries. _____ Barad's theory presents a paradigmatic shift in the scientific understanding of reality and the quantum world.

23 Mark for Review

Which choice completes the text with the most logical transition?

(A) For example,

(B) By contrast,

(C) As such,

(D) That said,

@DSATuz

Upper-atmospheric jets—phenomena whereby electrical discharges cause narrow cones of blue light to briefly burst upward from the tops of thunderclouds—have been observed reaching the ionosphere. The extreme altitudes involved (the ionosphere begins about 80 km above Earth) mark these gigantic jets as outliers; _____ the majority of jets reach heights of only 20 to 50 km.

24 Mark for Review

Which choice completes the text with the most logical transition?

(A) consequently,

(B) indeed,

(C) in addition,

(D) nevertheless,

@DSATuz

While researching a topic, a student has taken the following notes:

- In music theory, the term "key" refers to the set of musical notes that forms the foundation of a piece of music.
- In *Ideas Toward an Aesthetic of Music* (1806), German poet and composer Christian Schubart describes the moods of various musical keys.
- He describes the key of C minor as expressing "longing."
- In *Règles de Composition* (1682), French composer Marc-Antoine Charpentier undertakes the same task.
- He describes the key of C minor as "obscure and sad."
- "Don't Speak" by No Doubt (1996) is a song written in C minor.

25 Mark for Review

The student wants to compare how Schubart and Charpentier describe the mood of C minor. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Schubart and Charpentier each undertook the task of describing the mood of various keys, including C minor.

(B) Schubart describes the mood of C minor as expressing "longing," while Charpentier describes it as "obscure and sad."

(C) The song "Don't Speak" by No Doubt is written in the key of C minor, the mood of which has been described as expressing "longing."

(D) The key of C minor is one of various keys, which are sets of musical notes that form the foundations of pieces of music.

@DSATuz

- The Paralympic Games are a series of international sporting events involving athletes with an impairment.
- A gold medal indicates that an athlete finished in first place.
- Swimming has been an event at the Paralympics since 1960.
- Béatrice Hess competed as a Paralympic swimmer from 1984 to 2004.
- She won twenty-five medals.
- Of these medals, twenty were gold.

26 Mark for Review

Which choice most effectively uses information from the given sentences to emphasize how long Hess's sport has been a Paralympic event?

(A) An accomplished Paralympian with multiple gold medal wins, Béatrice Hess competed as a swimmer from 1984 to 2004.

(B) Béatrice Hess competed at the Paralympic Games, a series of international sporting events.

(C) Swimming, a sport in which Béatrice Hess took home twenty gold medals, has been an event at the Paralympic Games since 1960.

(D) During her career in swimming, which started in 1984, Béatrice Hess took home a total of twenty-five medals, twenty of which were gold.

@DSATuz

While researching a topic, a student has taken the following notes:

- A 2022 study analyzed geotagged social media posts to assess the effect of parks on happiness in 25 US cities.
- The happiness benefit of a city's parks was the numeric difference between the calculated happiness scores of posts made in parks and posts made elsewhere.
- A benefit above 0 indicates a positive effect.
- The park happiness benefit in San Francisco, California, was greater than 0.10.
- The park happiness benefit in Washington, DC, was greater than 0 but less than 0.10.
- The average happiness benefit of parks across all cities was 0.10.

27 Mark for Review

The student wants to make a generalization based on the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Geotagged social media posts were analyzed for expressed happiness in both San Francisco and Washington.

(B) Though it can vary from city to city, the overall effect of parks on happiness is a positive one.

(C) San Francisco's parks had a positive effect on happiness, but this was not true of Washington's parks, which had a below-average effect.

(D) By comparing the happiness scores of posts made in parks and posts made elsewhere, the study determined the effect of parks on happiness.

@DSATuz

1 Mark for Review

The table shows selected values from the function f.

x	$f(x)$
-1	$\frac{1}{50}$
0	1
1	50
2	$2{,}500$

Which of the following is the best description of function f?

(A) Increasing linear

(B) Decreasing linear

(C) Increasing exponential

(D) Decreasing exponential

@DSATUZ

2 Mark for Review

The equation $d = 37w + 100$ gives the total amount of money d, in dollars, that Janna plans to save w weeks after she starts to save. What is the total amount of money, in dollars, she plans to save in 10 weeks?

(A) 420

(B) 470

(C) 520

(D) 570

@DSATUZ

3 Mark for Review

What value of n is the solution to the equation $9n - 10 = 2n + 25$?

(A) 5

(B) 7

(C) 9

(D) 11

@DSATuz

4 Mark for Review

$$x^2 - 2 = 23$$

Which value is a solution to the given equation?

(A) -23

(B) -21

(C) -2

(D) 5

@DSA Tuz

5 Mark for Review

$$f(x) = 7x + 9$$

The function f gives the estimated height, in feet, of a poplar tree x years after its height was first measured. Which statement is the best interpretation of 9 in this context?

(A) The tree will be measured each year for 9 years.

(B) The tree is estimated to grow to a maximum height of 9 feet.

(C) The estimated height of the tree increased by 9 feet each year.

(D) The estimated height of the tree was 9 feet when it was first measured.

@DSATuz

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2

6 Mark for Review

The perimeter of a square can be expressed as $4(76)$ units. What is the length, in units, of a side of this square?

Answer Preview:

@DSATuz

7 Mark for Review

Sonja purchased a laptop in September 2018 when the laptop had a value of $1,000. Each month for the next 50 months, the value of the laptop decreased by 2% of the previous month's value. Which equation represents this situation, where v is the value of the laptop, in dollars, m is the number of months since Sonja purchased the laptop, and $m \leq 50$?

(A) $v = 2{,}018(0.98)^m$

(B) $v = 2{,}018(2)^m$

(C) $v = 1{,}000(0.98)^m$

(D) $v = 1{,}000(2)^m$

@DSATuz

8 Mark for Review

In triangle FGH, the measure of angle F is 63°. Which additional piece of information is sufficient to prove that triangle FGH is a right triangle?

(A) The length of $\overline{FH}$ is 63.

(B) The length of $\overline{GH}$ is 126.

(C) The measure of angle H is 27°.

(D) The sum of the measures of angles F, G, and H is 180°.

@DSATuz

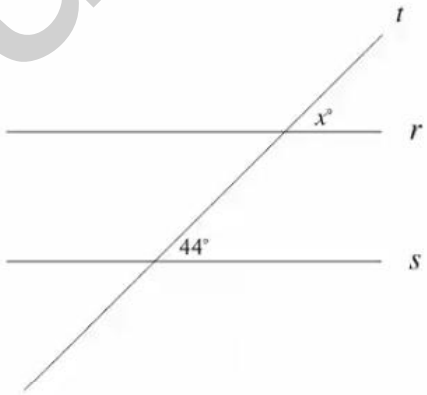
Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
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- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2

9 Mark for Review



Note: Figure not drawn to scale.

In the figure shown, line r is parallel to line s. What is the value of x?

Answer Preview:

@DSATuz

10 Mark for Review

$$11m = 7(n + p)$$

The given equation relates the positive numbers m, n, and p. Which equation correctly gives n in terms of m and p?

(A) $n = \frac{7p}{11m}$

(B) $n = \frac{11m}{7} - p$

(C) $n = 7(11m) + p$

(D) $n = 11m - 7 - p$

@DSA Tuz

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2

11 Mark for Review

A survey of the daily commuting times, in minutes, of seven different Chicago-based employees at a certain company yielded a data set consisting of the values 31, 13, 44, 32, 53, 22, and 44. A survey of the daily commuting times, in minutes, of seven different San Francisco–based employees at the same company yielded a data set consisting of the values 46, 31, 19, 33, 48, x, and 38. If the means of the two data sets are equal, what is the value of x?

Answer Preview:

@DSATuz

12 Mark for Review

Which expression is equivalent to $5xy + 30x^2y + 35xy^2$?

(A) $5xy(1 + 30x + 35y)$

(B) $5xy(1 + 6x + 7y)$

(C) $5xy(30x + 35y)$

(D) $5xy(6x + 7y)$

@DSATuz

13 Mark for Review

A survey was conducted in Madison to estimate the percent of adult residents who support the addition of streetlights at every corner in their town. Based on the survey, it is estimated that 40% of all adult residents of Madison support the addition of streetlights at every corner in their town, with an associated margin of error of 6%. Which of the following is the most appropriate conclusion about all adult residents of Madison?

(A) Plausible values for the percent of all adult residents of Madison who support the addition of streetlights at every corner in their town are between 34% and 46%.

(B) The addition of streetlights at every corner in the town is supported by exactly 34% of all adult residents of Madison.

(C) Plausible values for the percent of all adult residents of Madison who support the addition of streetlights at every corner in their town are either less than 34% or more than 46%.

(D) The addition of streetlights at every corner in the town is supported by more than 46% of all adult residents of Madison.

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14 Mark for Review

A plant nursery classifies a perennial flower species as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h, in centimeters, for a specific perennial flower species when fully grown. Which inequality represents the possible heights h, in centimeters, for a tall perennial flower species?

(A) $18 < h < 95$

(B) $41 < h < 89$

(C) $84 < h < 100$

(D) $110 < h < 158$

@DSATuz

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit

3.5	`3.5`
	`3.50`
	`7/2`
		`31/2`
		`3 1/2`

15 Mark for Review

$$(2x^2 + 3x - 3) - (7x^2 + 2x)$$

The given expression is equivalent to $-5x^2 + ax - 3$, where a is a constant. What is the value of a?

Answer Preview:

@DSATuz

16 Mark for Review

The kinetic energy K, in joules, of an object is given by the formula $K = \frac{1}{2}mv^2$, where m is the mass of the object, in kilograms, and v is the velocity of the object, in meters per second. If the kinetic energy of a certain object can be found by using the formula $K = 98v^2$, what is the mass of the object, in kilograms?

(A) 9,604

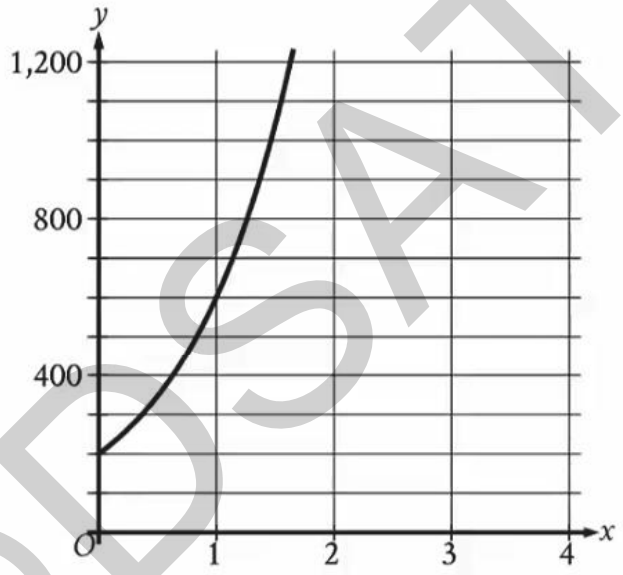
(B) 196

(C) 98

(D) 49

@DSATUZ

17 Mark for Review



The graph models the number of online newsletter subscribers at the end of every six-month period, where x is the number of six-month periods since the end of January 1995 and $0 \leq x \leq 4$. Which statement is the best interpretation of the point $(1, 600)$ in this context?

@DSATuz



The graph models the number of online newsletter subscribers at the end of every six-month period, where x is the number of six-month periods since the end of January 1995 and $0 \leq x \leq 4$. Which statement is the best interpretation of the point $(1, 600)$ in this context?

(A) The estimated number of online newsletter subscribers at the end of the first six-month period was 600.

(B) The estimated number of online newsletter subscribers increased every six months by 600 subscribers.

(C) The estimated number of online newsletter subscribers at the end of January 1995 was 600.

(D) The estimated number of online newsletter subscribers at the end of January 1996 was 600.

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Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit

3.5	`3.5`
	`3.50`
	`7/2`
		`31/2`
		`3 1/2`

18 Mark for Review

The table summarizes the number of objects in each group.

Group	Number of objects
A	325
B	64
C	611
D	96
Total	1,096

The number of objects in group C is $p\%$ of the number of objects in group A. What is the value of p?

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Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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Examples

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3.5	3.5 3.50 7/2	31/2 3 1/2

19 Mark for Review

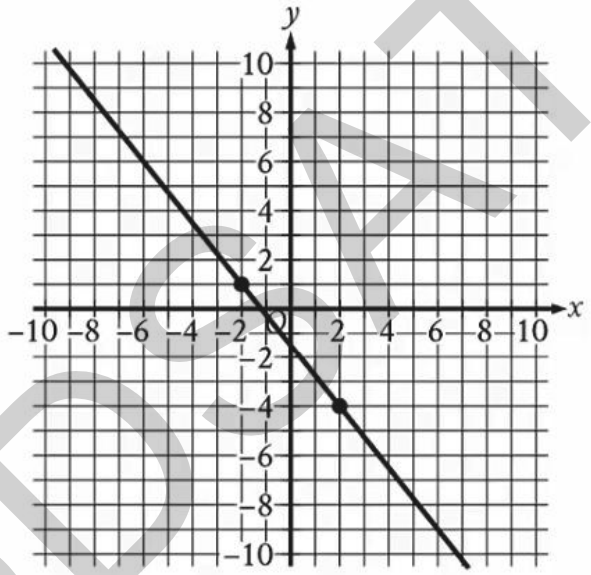
$$\frac{1}{4}(s+4t)=544$$
$$7s-8t=1{,}192$$

The solution to the given system of equations is (s,t). What is the value of $s-t$?

Answer Preview:

@DSATuz

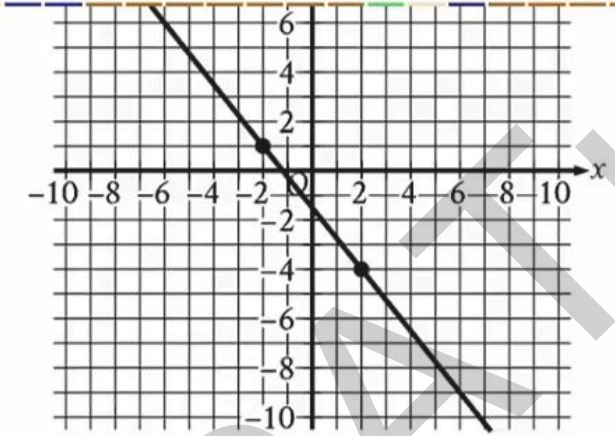
20 Mark for Review



Line k is shown in the xy-plane. Line j (not shown) is perpendicular to line k and passes through the point $(7, 15)$. What is the y-coordinate of the y-intercept of line j?

(A) $-\frac{47}{4}$

@DSATuz



Line k is shown in the xy-plane. Line j (not shown) is perpendicular to line k and passes through the point $(7, 15)$. What is the y-coordinate of the y-intercept of line j?

(A) $-\frac{47}{4}$

(B) $-\frac{3}{2}$

(C) $\frac{4}{5}$

(D) $\frac{47}{5}$

@DSAT.uz

Student-produced response directions

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Examples

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21 Mark for Review

x	$r(x)$
-41	-6
-21	0
-1	6

For the linear function r, the table shows three values of x and their corresponding values of $r(x)$. Function r is defined by $r(x) = ax + b$, where a and b are constants. What is the value of $a + b$?

Answer Preview:

@DSATuz

22 Mark for Review

$$y = -3x^2 - 40$$

$$y = qx - 37$$

In the given system of equations, q is a positive constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x?

(A) -77

(B) -43

(C) -6

(D) -1

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1 Mark for Review

For the linear function g, the graph of $y = g(x)$ in the xy-plane has a slope of 3 and passes through the point $(2, 28)$. Which equation defines g?

(A) $g(x) = 22x$

(B) $g(x) = 22x + 3$

(C) $g(x) = 3x$

(D) $g(x) = 3x + 22$

@DSA Tuz

2 Mark for Review

If the equations $-9x + y = 17$ and $9x + y = 17$ are graphed in the xy-plane, at what point (x, y) do the graphs intersect?

(A) $(0, 17)$

(B) $(0, 9)$

(C) $(9, 0)$

(D) $(17, 0)$

@DSATuz

3 Mark for Review

A baker has a total of 15 cups of flour and enough other ingredients to make banana pancakes and apple pancakes. The baker needs 1.25 cups of flour for a batch of banana pancakes and 0.75 cups of flour for a batch of apple pancakes. If the baker makes 8 batches of banana pancakes, what is the greatest number of whole batches of apple pancakes the baker can make?

(A) 6

(B) 7

(C) 8

(D) 9

@DSATuz

4 Mark for Review

What is the y-intercept of the graph of $y = 6(7)^{x+2}$ in the xy-plane?

(A) $(0, 6)$

(B) $(0, 84)$

(C) $(0, 294)$

(D) $(0, 1{,}764)$

@DSATuz

5 Mark for Review

$$x^2 - 11x - 3 = 0$$

What is a solution to the given equation?

(A) $\frac{11+\sqrt{(-11)^2-4(1)(-3)}}{2(-3)}$

(B) $\frac{11+\sqrt{(-11)^2-4(1)(-3)}}{2(1)}$

(C) $\frac{3+\sqrt{(-3)^2-4(1)(-11)}}{2(-3)}$

(D) $\frac{3+\sqrt{(-3)^2-4(1)(-11)}}{2(1)}$

@DSATuz

6 Mark for Review

$$\frac{2}{5}x + \frac{2}{3}y = 5$$

Which table gives three values of x and their corresponding values of y for the given equation?

(A)

x	y
1	$\frac{59}{15}$
2	$\frac{53}{15}$
4	$\frac{41}{15}$

(B)

x	y
1	$\frac{48}{5}$
2	$\frac{46}{5}$

@DSATuz

2	$\frac{46}{5}$
4	$\frac{42}{5}$

C

x	y
1	$\frac{42}{5}$
2	$\frac{39}{5}$
4	$\frac{33}{5}$

D

x	y
1	$\frac{69}{10}$
2	$\frac{63}{10}$
4	$\frac{51}{10}$

@DSATuz

7 Mark for Review

Ari examined a set of 86 plants. The lightest plant had a mass of 2.8 kilograms (kg) and the heaviest plant had a mass of 6.3 kg. Chihiro examined the same set of 86 plants and also an additional plant with a mass of 11.6 kg. Which of the following must be true about the set of plants that Ari examined and the set of plants that Chihiro examined?

(A) The standard deviation of the masses, in kg, of the plants that Ari examined is greater than the standard deviation of the masses, in kg, of the plants that Chihiro examined.

(B) The range of the masses, in kg, of the plants that Ari examined is greater than the range of the masses, in kg, of the plants that Chihiro examined.

(C) The median of the masses, in kg, of the plants that Ari examined is less than the median of the masses, in kg, of the plants that Chihiro examined.

(D) The mean of the masses, in kg, of the plants that Ari examined is less than the mean of the masses, in kg, of the plants that Chihiro examined.

@DSATUZ

8 Mark for Review

$$3x + 9y = 10$$
$$rx + 18y = 20$$

In the given system of equations, r is a constant. The system has infinitely many solutions. What is the value of r?

(A) 2

(B) 3

(C) 6

(D) 9

@DSATuz

9 Mark for Review

In triangle ABC, angle B is a right angle. The length of side AB is $10\sqrt{41}$ and the length of side BC is $24\sqrt{41}$. What is the length of side AC?

(A) $14\sqrt{41}$

(B) $26\sqrt{41}$

(C) $34\sqrt{41}$

(D) $\sqrt{34 \cdot 41}$

@DSA Tuz

10 Mark for Review

$$m(t) = -0.0274\left(\frac{t}{7}\right)^2 + 7.3873\left(\frac{t}{7}\right) + 75.032$$

The function m gives the predicted body mass $m(t)$, in kilograms (kg), of a male giraffe t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement "$m(140)$ is approximately equal to 212" in this context?

(A) The predicted body mass of a male giraffe was approximately 140 kg 212 days after it was born.

(B) The predicted body mass of a male giraffe was approximately 212 kg 140 days after it was born.

(C) The predicted body mass of a male giraffe was approximately 212 kg $\frac{140}{7}$ days after it was born.

(D) The predicted body mass of a male giraffe was approximately $\frac{140}{7}$ kg 212 days after it was born.

@DSATuz

11 Mark for Review

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 10 times the length of the corresponding side of $ABCD$. The area of $ABCD$ is 140 square units. What is the area, in square units, of $EFGH$?

(A) 14

(B) 100

(C) 1,400

(D) 14,000

@DSA Tuz

12 Mark for Review

A machine makes 8-inch, 5-inch, and 7-inch parts. During a certain day, the number of 8-inch parts that the machine makes is 3 times the number n of 5-inch parts, and the number of 7-inch parts is 12. During this day, the machine makes 100 parts total. Which equation represents this situation?

(A) $8(3n) + 5n + 7(12) = 100$

(B) $8n + 5n + 7n = 100$

(C) $3n + 12 = 100$

(D) $4n + 12 = 100$

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13 Mark for Review

If $2 + \frac{(1-x)}{2} = \frac{-2(1-x)}{3}$, what is the value of $1 - x$?

Answer Preview:

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14 Mark for Review

An object's speed, in meters per hour, is increasing at a rate of 1,490 meters per hour per second. Which of the following is closest to this rate in **feet per second squared**? (Use 1 meter = 3.28 feet.)

(A) 0.126

(B) 1.36

(C) 7.57

(D) 81.5

@DSA Tuz

15 Mark for Review

Which equation has no solution?

(A) $63x + 36 = 9(7x + 4)$

(B) $63x + 4 = 7x + 4$

(C) $63x + 360 = 9(7x + 16)$

(D) $63x + 51 = -63x - 51$

@DSATuz

16 Mark for Review

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(32) > 0$, $f(35) < 0$, and $f(38) > 0$, which of the following could be the value of $a + b$?

(A) 35

(B) 37

(C) 67

(D) 69

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17 Mark for Review

One gallon of sealant costs $13 and will cover 300 square feet of a surface. A deck has a total surface area of d square feet. Which equation represents the cost c, in dollars, of the sealant needed to cover the deck twice?

(A) $c = \frac{300d}{13}$

(B) $c = \frac{600d}{13}$

(C) $c = 13\left(\frac{d}{150}\right)$

(D) $c = 13\left(\frac{d}{300}\right)$

@DSA Tuz

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18 Mark for Review

$$\sqrt{k-x} = 42 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

Answer Preview:

@DSATuz

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19 Mark for Review

$$26z^{18} + bz^9 + 70$$

In the given expression, b is a positive integer. If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b?

Answer Preview:

@DSATuz

20 Mark for Review

In isosceles triangle ABC, sides AB and AC are congruent. Point D divides side BC such that the length of $\overline{BD}$ is $\frac{2}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD. If the length of $\overline{BE}$ is 14, what is the length of $\overline{CF}$?

(A) 28

(B) 35

(C) 49

(D) 98

@DSATUZ

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21 Mark for Review

In the xy-plane, a unit circle with center at the origin O contains point A with coordinates $(1, 0)$ and point B with coordinates $\left(-\frac{5}{\sqrt{89}}, \frac{8}{\sqrt{89}}\right)$. If the measure of angle AOB is w radians, what is the value of $\tan w$?

Answer Preview:

@DSATuz

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22 Mark for Review

$$x(kx - 112) = -8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

Answer Preview:

@DSATuz